

Fundamentals of Electrical and Electronics Engineering (FEEE)

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Lecture-21

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AC Power Analysis

1. Instantaneous Power
2. Average Power/ Effective Power / Real Power/Active Power
3. Apparent Power
4. Complex Power($\text{Real Power} + j \text{Reactive Power}$)

1. Instantaneous power (in watts)

$$p(t) = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2}V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

2. Average Power (in watts)

$$\begin{aligned} P &= \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) \\ &= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) \\ &= \text{Effective Power / Real Power/Active Power} \end{aligned}$$

6. Determine the average power generated by each source and the average power absorbed by each passive element in the circuit of Fig. below.

We apply mesh analysis as shown in Fig. (b). For mesh 1,

$$\mathbf{I}_1 = 4 \text{ A}$$

For mesh 2,

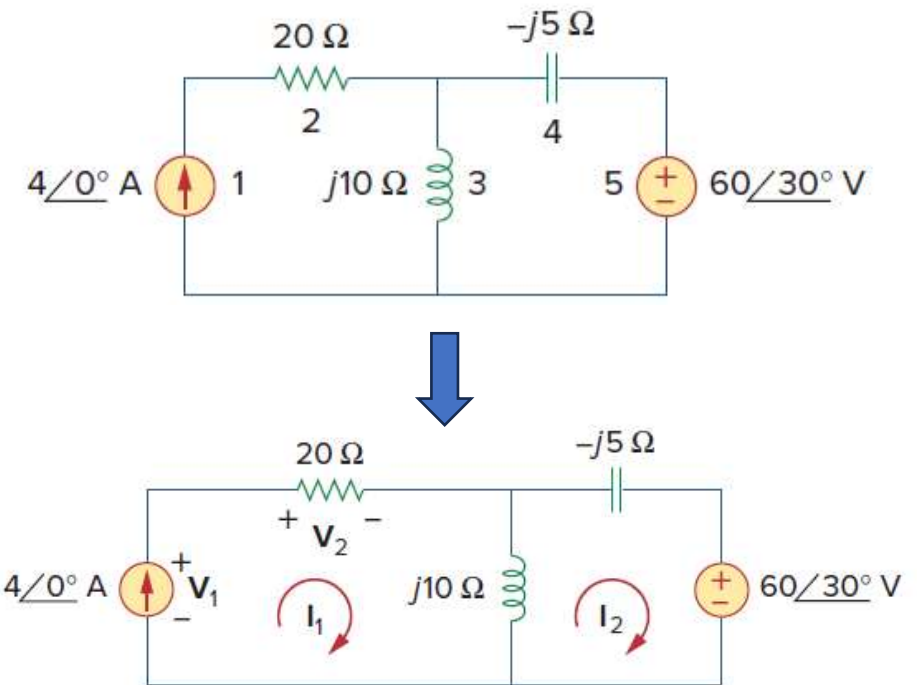
$$(j10 - j5)\mathbf{I}_2 - j10\mathbf{I}_1 + 60\angle 30^\circ = 0,$$

$$j5\mathbf{I}_2 = -60\angle 30^\circ + j40$$

$$\begin{aligned}\mathbf{I}_2 &= -12\angle -60^\circ + 8 \\ &= 10.58\angle 79.1^\circ \text{ A}\end{aligned}$$

the voltage across it is $60\angle 30^\circ \text{ V}$, so that the average power is

$$P_5 = \frac{1}{2} (60)(10.58) \cos(30^\circ - 79.1^\circ) = 207.8 \text{ W} \quad (1)\text{-Power absorbed}$$



For the resistor, the current through it is $\mathbf{I}_1 = 4\angle 0^\circ$ and the voltage across it is $20\mathbf{I}_1 = 80\angle 0^\circ$, so that the power absorbed by the resistor is

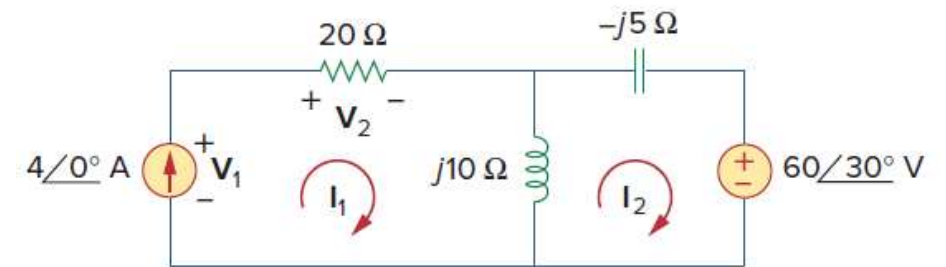
$$P_2 = \frac{1}{2} (80)(4) = 160 \text{ W} \quad (2)$$

For the current source, the current through it is $\mathbf{I}_1 = 4\angle 0^\circ$ and the voltage across it is

$$\begin{aligned} \mathbf{V}_1 &= 20\mathbf{I}_1 + j10(\mathbf{I}_1 - \mathbf{I}_2) \\ &= 80 + j10(4 - 2 - j10.39) \\ &= 183.9 + j20 = 184.984\angle 6.21^\circ \text{ V} \end{aligned}$$

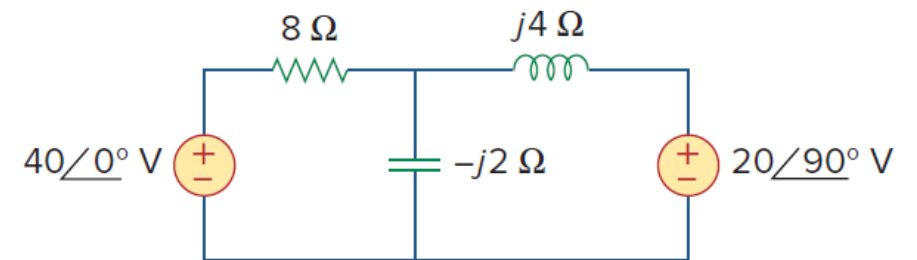
The average power supplied by the current source is

$$P_1 = \frac{1}{2} (184.984)(4) \cos(6.21^\circ - 0) = 367.8 \text{ W} \quad (3)$$



$$(1) + (2) = (3)$$

7. Calculate the average power of the five elements in the circuit of Fig. below.



Answer: 40-V Voltage source: 60 W; $j20$ -V Voltage source: 40 W; resistor: 100 W; others: 0 W.

8). A current $\mathbf{I} = 33 \angle 30^\circ$ A flows through an impedance $\mathbf{Z} = 40 \angle -22^\circ \Omega$. Find the average power delivered to the impedance.